VUT AI SOLUTIONS / BUSINESS ANALYSIS 3.2

StockSense AI

Sales forecasts and clearer stock orders for Sunrise Mini Market

Fictional retailer using anonymous Kaggle Store 1 and Item 1–5

The problem

A small retailer must choose an order quantity before knowing next week's sales. Guessing can leave too little or too much stock. StockSense gives the manager a consistent estimate to check before ordering.

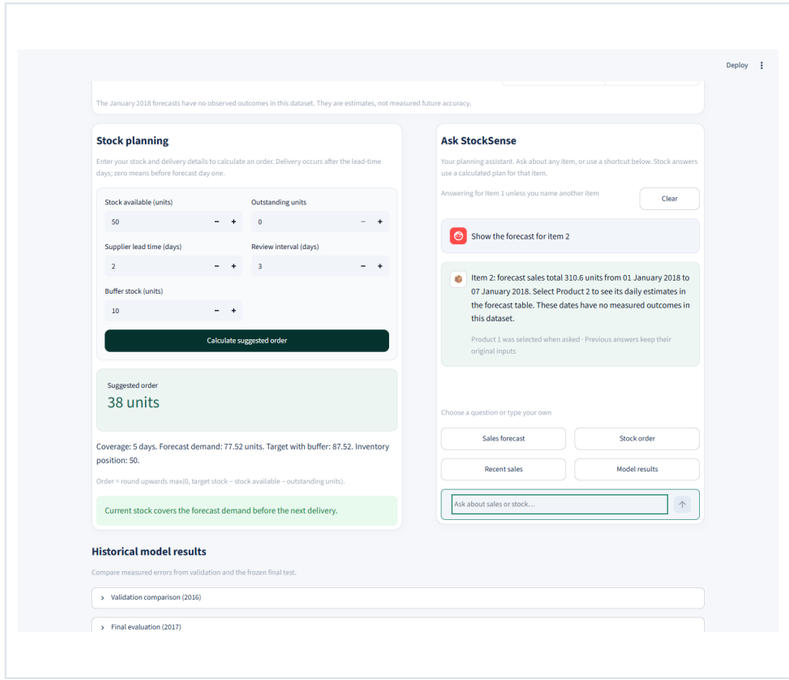
What the app does

1. Shows seven daily sales estimates for a selected product.

2. Combines the forecast with stock and delivery inputs.

3. Shows a suggested order and warns about shortages before delivery.

Working prototype



Product 1 example: 38 units suggested with the entered stock values

Measured forecast error

2017 final test: daily MAE in units. Lower is better.

Weekly benchmark — **6.273**

Small LSTM — **5.411**

Random forest — **4.812**

Random forest was chosen using 2016 validation results. Its 2017 daily MAE was 23.3% lower than the weekly benchmark. The LSTM had the lowest overall weekly-total error.

Scope and techniques: 9,130 sales records from 2013–2017; one store and five anonymous items. The app compares a weekly benchmark, random forest and a small LSTM. Its text assistant classifies five request types and retrieves the displayed results.

Limit: Stock inputs are demonstration values. January 2018 has no observed outcomes in this dataset. No real stockout or cost savings are claimed.

Source: Kaggle Store Item Demand Forecasting Challenge; frozen StockSense evaluation, 12 October 2026.